



A Hybrid Neural-Statistical Approach to Fake News Detection: Integrating TF-IDF and BERT Embeddings for Social Media Analysis

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Abstract.

Purpose: As the spread of fake news through social networks accelerates, the need to develop a reliable framework that can detect them increases. In this context, this dissertation proposes a framework that combines statistical and semantic textual representations.

Methods: In this study, we used the WELFake index, which contains 71 537 posts. During text preparation, case normalization and noise removal were performed. Two feature extraction methods were used: TF-IDF vectorization was used to determine the importance of keywords and lexical structures, and BERT Base Uncased was used to extract 768-dimensional contextual embeddings. These two sets of features were horizontally concatenated to generate a 10 768-dimensional unique feature representation. Classification was performed using a Random Forest classifier, which is based on this feature representation.

Result: Experimental results show that the proposed hybrid method is efficient in distinguishing between true and false messages. On the entire dataset, the overall accuracy of the sample was 87.4%, precision was 85.66%, recall was 90.94%, F1 score was 88.22%, and AUC was 0.9467. This indicates that the method combining statistical and semantic properties provides high classification efficiency in the problem of false message detection.

Novelty: Although existing methods are based only on statistical features or Transformer-based word insertions, this study integrates TF-IDF and BERT representations into a single feature fusion structure. Through the proposed method, the model and examples can capture key word-level and contextual semantic information, thus improving the effectiveness of false information detection.

Keywords: Fake News Detection, BERT, TF-IDF, Feature Fusion, Random Forest, Natural Language Processing.

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INTRODUCTION

The rapid development of social media has changed the way people access and share information. Platforms such as Twitter, Facebook, and Instagram allow for the rapid dissemination of information to a wide range of users. However, this ease of access has also facilitated the spread of misinformation and fake news. In situations such as the COVID-19 pandemic, misinformation has become widespread and has influenced public opinion, making the detection of misinformation a major research topic [1], [6], [14].

Various solutions have been proposed to solve this problem. Previous research has used deep neural network technologies, such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), to classify message content using textual features [3]. Subsequent research has developed CNN-LSTM and Graph Neural Networks (GNN) models to understand context and social interactions [10], [2]. These approaches have produced significant results and demonstrated the power of AI in detecting fake news.

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With recent advances in the field of natural language processing (NLP), the introduction of transformer-based models has become active. BERT and its variations have demonstrated superiority in the ability to capture contextual meaning more effectively than existing machine learning methods [1], [5], [7]. In addition, research is underway to explore advanced transformer architectures such as XLNet and RoBERTa, as well as to develop explainable artificial intelligence technologies to increase the clarity of models and user confidence [17], [16], [18].

Many studies have reported high classification accuracy, but most of them were based only on textual statistical features or contextual embeddings. Recent studies by Aprilio et al. [8] and Okoro et al. [20] suggest that a combination of various feature extraction methods can improve the effectiveness of the model. However, research on detecting false information by integrating TF-IDF and BERT is still rare. Therefore, this study proposes a hybrid feature fusion method that combines TF-IDF and BERT representations to improve the efficiency of classifying false information in social media data. The main goal is to include both keyword-level information and contextual semantics in a single classification system.

METHODS

The flowchart in Figure 1 demonstrates the general flow of the architecture proposed for fake news detection. The hybrid architecture involves data preparation, text preprocessing, feature extraction using TF-IDF and BERT, feature fusion, random forest classification, and performance evaluation. The idea of using the two types of representation is based on adding more information from texts to increase the accuracy of discrimination between real and fake news.

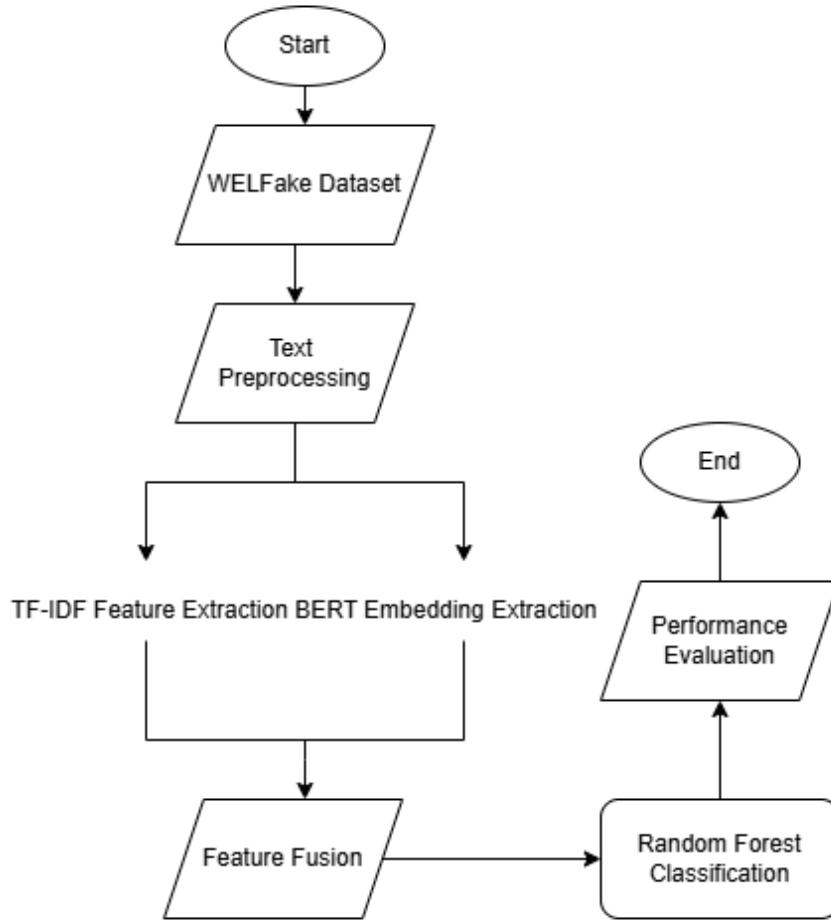


Figure 1. Research Workflow

In this study, the WELFake index was used, consisting of 71,537 messages collected from online news portals and social media. The WELFake index consists of 36,509 fake messages and 35,028 true messages. The training and testing indexes were divided 80:20, using 57,229 training examples and 14,308 testing examples. Such referential indexes have been widely used in fake news detection research due to the different linguistic structures of true and false data [1], [6], [14].

Before feature extraction, several pre-processing steps were performed to improve the quality and consistency of the dataset. The titles and contents of the articles were merged into a single text field. Then, all characters were converted to lowercase, URLs were removed, non-alphabetic symbols were removed, and excess whitespace was normalized. These pre-processing procedures reduce textual noise while preserving the semantic information necessary for classification. Similar pre-processing procedures have been used in previous studies on the detection of false information, based on Transformer algorithms [8], [9], [15].

The first feature extraction method used in this study is the Term Frequency–Inverse Document Frequency (TF-IDF) method. This method estimates the statistical importance of terms based on their frequency in documents and their rarity in the corpus. In this study, the TF-IDF method was constructed using up to 10,000 features with monolingual and bilingual expressions. The TF-IDF scores are calculated using Equation (1).

$$TFIDF(t, d) = TF(t, d) \times \log \left(\frac{N}{DF(t)} \right)$$

Here, $TF(t,d)$ represents the frequency of the term "t" in the document "d", $DF(t)$ represents the number of documents containing the term "t", and NN represents the total number of documents in the corpus. TF-IDF remains an effective method for identifying lexical patterns and distinguishing key words often associated with misinformation content [10], [18], [21].

The BERT (Bidirectional Encoder Representations from Transformers) algorithm was used to capture contextual semantic information. The BERT Base Uncased model was pre-trained to generate 768-dimensional word embeddings from the textual data. Unlike the traditional "bag-of-words" method, BERT considers bidirectional contextual relationships between words, allowing for a more detailed understanding of the semantics of sentences. Due to the computational cost of generating word embeddings using the Transformer algorithm, a representative set consisting of 2000 training examples and 1000 testing examples was used for embedding extraction. Previous research has shown that BERT-based architectures effectively mimic semantic context, consistently achieving superior performance in disinformation detection tasks [1], [6], [9], [11], [13].

After extracting statistical and semantic features, feature-level fusion was performed through horizontal linking. A TF-IDF vector containing ten thousand features was combined with a 768-dimensional BERT embedding to generate a synthetic representation containing ten thousand seven hundred sixty-eight features for each message. The fusion process is described in Equation (2).

$$X_{fusion} = X_{TFIDF} || X_{BERT}$$

Here, $||$ represents the concatenation operation. The resulting hybrid representation integrates the keyword information from TF-IDF with the contextual semantic information obtained by BERT. Similar hybrid methods are known to improve classification performance by combining different perspectives of textual information [10], [19].

The combined feature vectors were used to train a Random Forest classifier. Random Forest is a learning algorithm based on sequences that constructs multiple decision trees and integrates their outputs to generate a final prediction. This classifier was chosen because of its robustness to high-dimensional feature spaces and its ability to reduce overfitting by sequence-based decision. Due to these properties, Random Forest is suitable for hybrid feature representations generated from TF-IDF and BERT features [20], [22].

To evaluate the effectiveness of the proposed structure, several performance measures were used, including accuracy, precision, recall, F1 score, and AUC (area under the curve). Accuracy measures the overall performance of the classification and is calculated using equation (3).

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Precision estimates the proportion of correctly predicted positive samples and is calculated by equation (4).

$$Precision = \frac{TP}{TP+FP}$$

The recall rate measures the ability of a sample to accurately identify positive cases and is calculated using equation (5).

$$Recall = \frac{TP}{TP+FN}$$

The F1 point represents the harmonic mean of precision and recall and is calculated by equation (6).

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Here, TPTPTP, TNTNTN, FPFPPF, and FNFNFN represent the true positive, true negative, false positive, and false negative values, respectively. These indices are commonly used in false information detection research because they can fully evaluate the classification efficiency and model reliability [7], [20], [22].

RESULT AND DISCUSSION

Dataset Characteristics

The WELFake dataset used in this study consists of 71,537 messages, of which 36,509 are fake messages and 35,028 are legitimate articles. The distribution of the two classes is relatively even, which reduces the risk of misclassification and provides a suitable environment for evaluating fake message detection models. The dataset was divided into a training set and a testing set in a ratio of 80:20, with 57,229 data points in the training set and 14,308 data points in the testing set. In the process of incorporating the BERT example, 2,000 data points of the training dataset and 1,000 data points of the test dataset were randomly chosen so as to decrease computational complexity without sacrificing the needed text diversity.

Table 1. Distribution of WELFake dataset

Category	Description	Value
Dataset Size	WELFake Dataset	71,537 Articles
Fake News	Label = 1	36,509
Real News	Label = 0	35,028
Training set	80% Split	57,229
Testing set	20% Split	14,308
BERT Training Subset	Embedding Generation	2,000
BERT Testing Subset	Embedding Generation	1,000
TF-IDF Features	Unigram + Bigram	10,000
BERT Features	BERT Base Uncased	768
Fused Features	TF-IDF + BERT	10,768

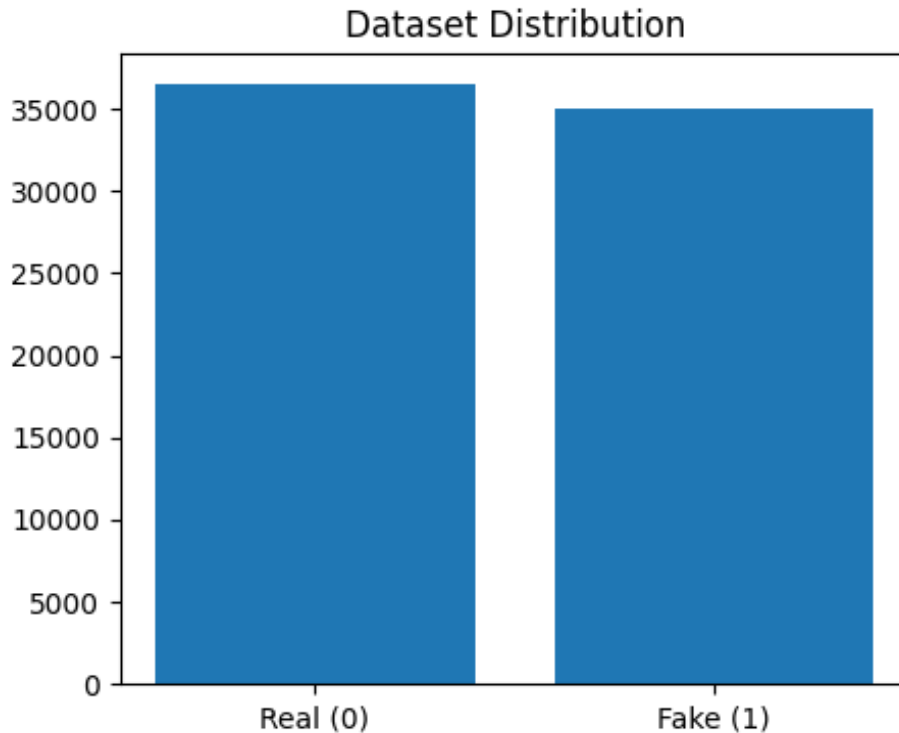


Figure 2. Distribution of false and true messages in the WELFake dataset

As we can see in Figure 2 below, the data set exhibits almost an equal distribution between the true and false messages. This is advantageous because it enables the classifier to develop a model of both sides of the distribution without any one side taking dominance.

Feature Extraction and Fusion Analysis

The technique used in this paper is a hybrid fusion method that combines statistically generated feature representations with semantically generated representations. TF-IDF generates 10,000 lexical features, and BERT generates 768-dimensional contextual embeddings. These two features are combined through a horizontal join technique to generate a 10,768-dimensional hybrid feature vector.

Table 2. Feature Extraction Statistics

Feature Representation	Dimension
TF-IDF Features	10,000
BERT Embeddings	768
Hybrid Feature Vector	10,768

The hybrid method allows for the inclusion of important word patterns, importance, and contextual semantic information. While TF-IDF studies word frequency patterns related to true/false information, BERT studies the relationships between words and sentences within context. In this way, hybrid classifiers use additional information not included in feature extraction.

Classification Performance

These feature vectors were then fed to the Random Forest algorithm. The performance of the proposed architecture was measured using the accuracy, precision, recall, F1 score, and AUC (Area Under the Curve).

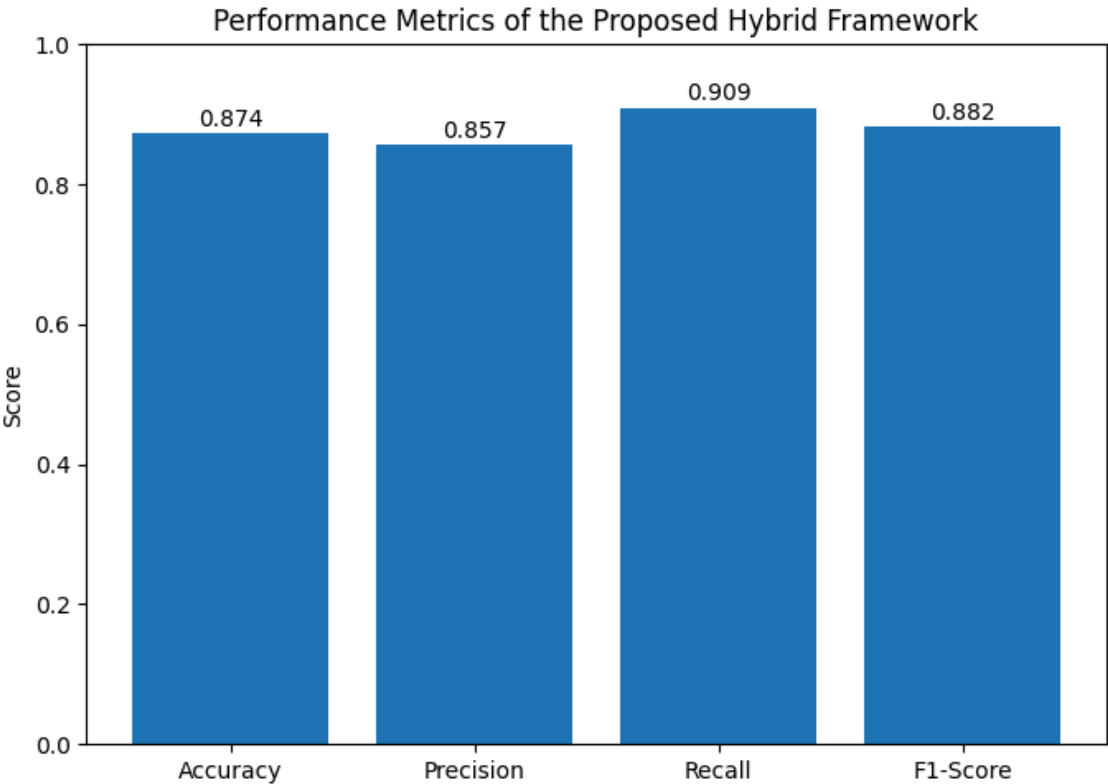


Figure 3. Performance Metrics of the proposed Hybrid Framework

Figure 3 shows the performance measurement results of the proposed hybrid method. The model achieved an accuracy of 87.40%, which means that most of the messages were correctly classified. The accuracy of 85.66% indicates that a significant number of false positives predicted by the model were correctly classified. Furthermore, the high recall value of 90.94% demonstrates the model’s strong ability to

identify false positives. The F1 value of 88.22% represents a balance between accuracy and recall, showing good discriminative power with an AUC of 0.9467.

Confusion Matrix Analysis

To investigate the proposed classification structure in more detail, we generated a confusion matrix. This matrix represents the number of correctly classified and incorrectly classified examples for each class, illustrating the accuracy of the prediction results.. Unlike general accuracy, this analysis allows for a more in-depth examination of classification errors and performance per class, making it a widely used evaluation tool in disinformation detection research [12], [21], [22].

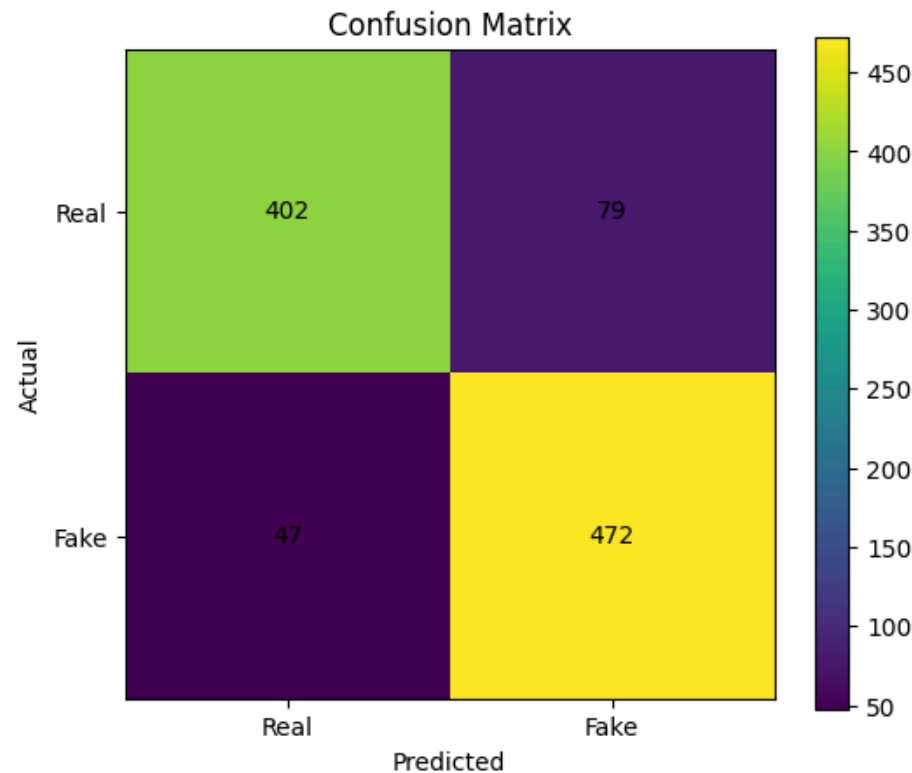


Figure 4. Confusion matrix of the proposed hybrid structure.

As shown in Figure 4, most of the messages were accurately classified into their categories, indicating that the hybrid feature representation effectively captured the relevant lexical and semantic properties of the input text. The number of accurate predictions far exceeded the number of classification errors, demonstrating the effectiveness of the proposed fusion strategy of TF-IDF and BERT. Similar improvements in misinformation detection have been observed using a hybrid approach combining statistical and contextual representations [8], [18].

The confusion matrix also confirms the performance measures presented in the previous section. The recall ratio of 90.94% indicates that the model successfully identified most of the false messages, which is especially important in misinformation detection tasks. This is because if false content is not detected, misleading information can spread. Previous studies using transformer-based architectures have also shown that high recall is a significant indicator of the effectiveness of a model in classifying false messages [1], [5], [14].

Although a limited number of false positives and misclassifications were observed, given the complex linguistic structure and contextual ambiguity of web messages, such errors are expected in the true detection of misinformation. Some misinformation articles may be stylistically very similar to legitimate articles, while some legitimate articles may contain sensational language, close to misinformation. Similar

issues have been addressed in research on BERT and deep learning-based misinformation detection systems [6], [10], [16].

Overall, the confusion matrix analysis revealed that the proposed hybrid structure provides balanced classification performance for both classes. The high recall, fairly high accuracy, and low classification error rate demonstrate the robustness of the model and confirm its suitability for real-world applications of false information detection. These results are consistent with previous studies that show that integrating lexical and semantic attributes can improve classification reliability and generalization performance [8], [18], [19].

ROC Curve and AUC Analysis

To more accurately assess the discriminative power of the proposed hybrid architecture, we performed a ROC (Return on Receiver) curve analysis. The ROC curve is a graph of the True Positive Rate (TPR) and False Positive Rate (FPR) used to evaluate the effectiveness of a classifier under various discrimination thresholds. Unlike precision-based measures, ROC curves allow for the prediction of the effectiveness of a classifier, regardless of the threshold.

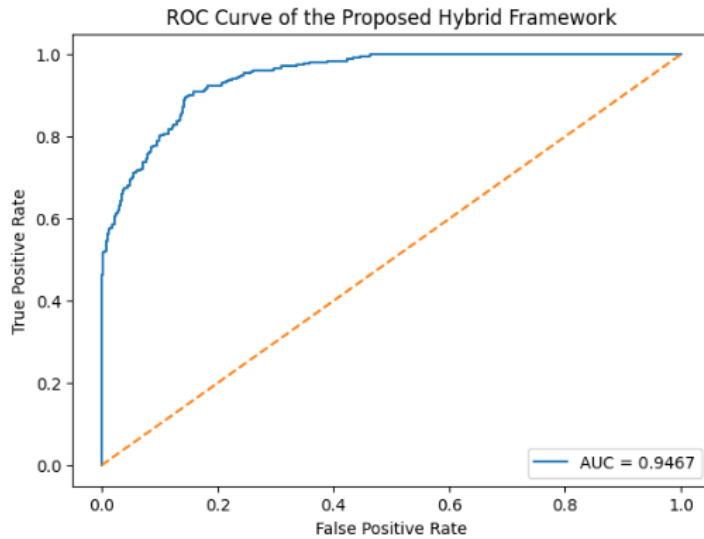


Figure 5. ROC Curve of the Proposed Hybrid Framework

As shown in Figure 5, the ROC curve is located near the top left of the graph, indicating a good balance between sensitivity and specificity. The proposed method has an Area Under the Curve (AUC) value of 0.9467, which is close to the maximum value of 1.0. In classification tasks, an AUC value of 0.90 or higher is generally considered to represent the best performance. This implies a high probability that the classifier can accurately distinguish between fake and legitimate messages.

The high AUC value obtained in this study demonstrates the effectiveness of integrating the TF-IDF function with the BERT contextual functions. While TF-IDF provides lexical information about word frequency and document relevance, BERT captures semantic relationships and contextual meaning within the text. By combining these complementary feature representations, the Random Forest classifier can make more reliable predictions for various decision thresholds.

In addition to this, the ROC (Receiver Operating Characteristic for Operation) is the technique used to check the accuracy, precision, recall, and F1 scores given in the previous section. As we can see, there is consistency between the measures used for the evaluation which shows that the efficient classification model is consistent in its performance and is not based on one measure to prove its efficiency. Thus, the results from the AUC values prove the effectiveness of the proposed method.

CONCLUSION

In this study, a mixed approach is developed using TF-IDF and BERT word embedding features. The aim of this study is to develop the efficiency of detecting the fake news by using statistical features and contextual semantic representation of key words. The proposed approach involves the integration of 10,000 TF-IDF features with 768-dimensional BERT embeddings via WELFake index and applying classification with the help of Random Forest classifier. The experimental results reveal very high efficiency since the following metrics were achieved: accuracy of 87.40%, precision of 85.66%, recall of 90.94%, F1-score of 88.22%, and AUC of 0.9467. Thus, the obtained results indicate the high efficiency of lexical and semantic features in terms of being capable of discriminating fake news from the legitimate one. Moreover, this study indicates that the combination of existing statistical approaches with a transformer-based language model leads to high classification efficiency in comparison with the approaches based on single feature representation. In addition, there is a possibility to increase the efficiency of detecting fake news in future research by developing large-scale transformer-based architectures, other cluster classifiers, as well as multimodal approaches with regard to the integration of images, users' behaviour, and social network data.

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